

Informatica® Data Replication
9.8.0

Installation and Upgrade Guide

Informatica Data Replication Installation and Upgrade Guide
9.8.0
December 2018

© Copyright Informatica LLC 2017, 2018

This software and documentation are provided only under a separate license agreement containing restrictions on use and disclosure. No part of this document may be reproduced or transmitted in any form, by any means (electronic, photocopying, recording or otherwise) without prior consent of Informatica LLC.

U.S. GOVERNMENT RIGHTS Programs, software, databases, and related documentation and technical data delivered to U.S. Government customers are "commercial computer software" or "commercial technical data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, duplication, disclosure, modification, and adaptation is subject to the restrictions and license terms set forth in the applicable Government contract, and, to the extent applicable by the terms of the Government contract, the additional rights set forth in FAR 52.227-19, Commercial Computer Software License.

Informatica, the Informatica logo, and PowerCenter are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html. Other company and product names may be trade names or trademarks of their respective owners.

Portions of this software and/or documentation are subject to copyright held by third parties. Required third party notices are included with the product.

The information in this documentation is subject to change without notice. If you find any problems in this documentation, report them to us at infa_documentation@informatica.com.

Informatica products are warranted according to the terms and conditions of the agreements under which they are provided. INFORMATICA PROVIDES THE INFORMATION IN THIS DOCUMENT "AS IS" WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING WITHOUT ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND ANY WARRANTY OR CONDITION OF NON-INFRINGEMENT.

Publication Date: 2018-12-07

Table of Contents

Preface 4

Informatica Resources 4

- Informatica Network 4
- Informatica Knowledge Base 4
- Informatica Documentation 4
- Informatica Product Availability Matrixes 5
- Informatica Velocity 5
- Informatica Marketplace 5
- Informatica Global Customer Support 5

Chapter 1: Installing Data Replication 6

Installation Overview 6

Before You Install 6

System Requirements 7

- AIX System Requirements 7
- HP-UX System Requirements 7
- Linux System Requirements 8
- Solaris System Requirements 8
- Windows System Requirements 9
- Source Database Versions 9
- Target Database Versions 10

Installing Data Replication 11

Post-Installation Tasks 12

- Installing the Java Runtime Environment 12
- Defining the DBSYNC_HOME Environment Variable 13
- Adding DBSYNC_HOME Libraries to the Library Path Environment Variable on Linux and UNIX 13
- Verifying the Environment Variables on Linux and UNIX 13
- Setting Limits for System Resources on Linux and UNIX 14
- Entering the License Key 14
- JDBC Drivers 14
- ODBC Drivers 16
- Configuring the DataDirect ODBC Drivers and Driver Manager 18

License Expiration 19

Chapter 2: Upgrading Data Replication 20

Overview 20

Upgrading Data Replication 20

Index 24

Preface

This *Informatica Data Replication Installation and Upgrade Guide* describes how to install or upgrade the Informatica® Data Replication product.

Informatica Resources

Informatica Network

Informatica Network hosts Informatica Global Customer Support, the Informatica Knowledge Base, and other product resources. To access Informatica Network, visit https://network.informatica.com.

As a member, you can:

- Access all of your Informatica resources in one place.
- Search the Knowledge Base for product resources, including documentation, FAQs, and best practices.
- View product availability information.
- Review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

Informatica Knowledge Base

Use the Informatica Knowledge Base to search Informatica Network for product resources such as documentation, how-to articles, best practices, and PAMs.

To access the Knowledge Base, visit https://kb.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

Informatica Documentation

To get the latest documentation for your product, browse the Informatica Knowledge Base at https://kb.informatica.com/_layouts/ProductDocumentation/Page/ProductDocumentSearch.aspx.

If you have questions, comments, or ideas about this documentation, contact the Informatica Documentation team through email at infa_documentation@informatica.com.

Informatica Product Availability Matrixes

Product Availability Matrixes (PAMs) indicate the versions of operating systems, databases, and other types of data sources and targets that a product release supports. If you are an Informatica Network member, you can access PAMs at https://network.informatica.com/community/informatica-network/product-availability-matrices.

Informatica Velocity

Informatica Velocity is a collection of tips and best practices developed by Informatica Professional Services. Developed from the real-world experience of hundreds of data management projects, Informatica Velocity represents the collective knowledge of our consultants who have worked with organizations from around the world to plan, develop, deploy, and maintain successful data management solutions.

If you are an Informatica Network member, you can access Informatica Velocity resources at http://velocity.informatica.com.

If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

Informatica Marketplace

The Informatica Marketplace is a forum where you can find solutions that augment, extend, or enhance your Informatica implementations. By leveraging any of the hundreds of solutions from Informatica developers and partners, you can improve your productivity and speed up time to implementation on your projects. You can access Informatica Marketplace at https://marketplace.informatica.com.

Informatica Global Customer Support

You can contact a Global Support Center by telephone or through Online Support on Informatica Network.

To find your local Informatica Global Customer Support telephone number, visit the Informatica website at the following link:
http://www.informatica.com/us/services-and-training/support-services/global-support-centers.

If you are an Informatica Network member, you can use Online Support at http://network.informatica.com.

CHAPTER 1

Installing Data Replication

This chapter includes the following topics:

- Installation Overview, 6
- Before You Install, 6
- System Requirements, 7
- Installing Data Replication, 11
- Post-Installation Tasks, 12
- License Expiration, 19

Installation Overview

Install Data Replication on all of the systems where you plan to run Data Replication components. Prior to installation, ensure that these systems meet operating system and hardware requirements of the product.

Note: You can run the Data Replication Console only on 64-bit Windows systems.

Informatica provides a platform-specific .zip file or .tar.gz file for installing Data Replication on a supported Linux, UNIX, or Windows system. To install Data Replication, you simply unzip or untar the installation file.

On each system where you install Data Replication, define the DBSYNC_HOME environment variable and point it to the root Data Replication directory. To use Data Replication, you must enter the valid license key that you received from Informatica.

Before You Install

Before you begin installation, verify that the following requirements are met:

- The system meets Data Replication system requirements.
- The system has enough free disk space. For the on-disk queue files that contain the change data, the amount of space depends on the transaction volume. As a general rule, 500 MB is sufficient. For a Data Replication Console client that is installed on a system separate from the source, at least 200 MB is required.
- You have a valid license key and an installation file for your platform.
- On AIX, IBM openssl 1.0.1 is installed.

System Requirements

Verify that the system where you plan to install Data Replication meets the product operating system and hardware requirements.

Tip: For the most current information, see the Data Replication Product Availability Matrix on the Informatica My Support Portal at http://mysupport.informatica.com.

AIX System Requirements

You can run Data Replication on the following versions of AIX 64-bit Power operating systems:

- 6.1 TL8
- 7.1 TL2
- 7.2 TL2

The following table shows the recommended processor type, memory, and disk space:

System Characteristic	Recommended
Processor	64-bit Power PC
RAM	1 GB or greater
Swap space	1 GB or greater
Disk space	500 MB

Verify that the bos.rte.streams package is installed on your system. Data Replication requires this package to run the Server Manager. If the bos.rte.streams package is not installed on your system, download and install it manually. To install the bos.rte.streams package, run the following command from the directory to which you decompressed the package:

```
smit install_latest
```

HP-UX System Requirements

You can run Data Replication on the 64-bit HP-UX 11.31 Itanium operating system.

The following table shows the recommended processor type, memory, and disk space:

System Characteristic	Recommended
Processor	Itanium, 64-bit
RAM	1 GB or greater
Swap space	1 GB or greater
Disk space	500 MB

Linux System Requirements

You can run Data Replication on the following versions of 64-bit Linux operating systems:

- Red Hat Enterprise Linux 6.5
- Red Hat Enterprise Linux 6.6
- Red Hat Enterprise Linux 6.7
- Red Hat Enterprise Linux 6.8
- Red Hat Enterprise Linux 6.9
- Red Hat Enterprise Linux 7
- Red Hat Enterprise Linux 7.1
- Red Hat Enterprise Linux 7.2
- Red Hat Enterprise Linux 7.3
- Red Hat Enterprise Linux 7.4
- SUSE Linux Enterprise Server 11 SP4
- SUSE Linux Enterprise Server 12 SP3

The following table shows the recommended processor type, memory, and disk space:

System Characteristic	Recommended
Processor	64-bit Opteron, EM64T
RAM	1 GB or greater
Swap space	1 GB or greater
Disk space	500 MB free space

Solaris System Requirements

You can run Data Replication on the following versions of Solaris 64-bit SPARC operating systems:

- 10
- 11

The following table shows the recommended processor type, memory, and disk space:

System Characteristic	Recommended
Processor	SPARC II, IIi, or III
RAM	1 GB or greater
Swap space	1 GB or greater
Disk space	500 MB free space

Windows System Requirements

You can run Data Replication on the following 64-bit Windows operating systems:

- Windows Server 2008 R2
- Windows Server 2012 R2
- Windows Server 2016
- Windows 10

The following table shows the recommended processor type, memory, and disk space:

System Characteristic	Recommended
Processor	64-bit Pentium 4 or later
RAM	1 GB or greater
Disk space with NTFS	500 MB free space
New Technology File System (NTFS)	Available

Source Database Versions

The following table identifies the versions of each source type that Data Replication supports:

Source Type	Supported Versions
DB2 for Linux, UNIX, and Windows	- 10.1 - 10.5 - 11.1
Microsoft SQL Server Standard Edition and Enterprise Edition[1]	- SQL Server 2012 - SQL Server 2014 - SQL Server 2016 - SQL Server 2017
MySQL Enterprise Edition[2]	- 5.7.*x*
Oracle and Amazon RDS for Oracle	- 11*g* R2 - 12*c* R1[3] - 12*c* R2[3] - 18*c*

1. You can run the Extractor for Microsoft SQL Server sources only on Linux or Windows operating systems. If you use SQL Server 2017 on Linux, the Extractor can process online transaction logs and archived transaction logs on the SQL Server system. If you use earlier SQL Server versions that run on Windows only, the Extractor can run on Linux and process only the archived transaction logs that were copied to the Linux system.

2. You can run the Extractor for MySQL sources only on Linux or Windows operating systems.

3. Data Replication supports Oracle 12*c* features with the following limitations:
- If the source database uses Oracle In-Database Archiving, Data Replication does not replicate the ROW ARCHIVAL clause and does not replicate DML or DDL operations for the hidden system column ORA_ARCHIVE_STATE.
- If a DDL statement for a source column contains the IDENTITY clause, the Applier executes the equivalent DDL statement on the target but without the IDENTITY clause.
- To capture changes in an Oracle multitenant environment, the source tables must reside in a single pluggable database (PDB) within a multitenant container database (CDB).

Target Database Versions

The following table identifies the versions of each target type that Data Replication supports:

Target Type	Supported Versions
Amazon Redshift[1]	-
Apache Kafka[2]	- 0.11 - 1.0 - 1.1 - 2.0
Cloudera	- 5.4 - 5.5 - 5.6 - 5.7 - 5.8 - 5.9 - 5.10 - 5.11 - 5.12 - 5.13 - 5.14
DB2 for Linux, UNIX, and Windows	- 10.1 - 10.5 - 11.1
Greenplum	- 4.3 - 5.*x*
Hortonworks	- 2.4 - 2.5 - 2.6
MemSQL	- 6.0
Microsoft SQL Server Standard Edition and Enterprise Edition	- SQL Server 2012 - SQL Server 2014 - SQL Server 2016 - SQL Server 2017
MySQL Enterprise Edition	- 5.7.*x* - 8.0
Netezza	- 7. *x*
Oracle	- 11*g* R2 - 12*c* R1 - 12*c* R2 - 18*c*
PostgreSQL	- 9.4 - 9.5 - 9.6 - 10.*x*

Target Type	Supported Versions
Teradata	- 15.10 - 16.00 - 16.10 - 16.20
Vertica	- 7.*x* - 8.*x* - 9.*x*

1. For Amazon Redshift targets, the Applier and InitialSync are not supported on AIX, HP-UX, and Solaris operating systems.
2. The Applier for Apache Kafka targets is supported on Linux and Windows operating systems only.
3. Data Replication does not use Scala binary code but does require the Kafka libraries from the binary download to be present.

Installing Data Replication

Install Data Replication on each system where you will run a Data Replication component such as the Extractor, Applier, Server Manager, InitialSync, and Data Replication Console. Use this procedure for a new installation.

To upgrade an existing Data Replication installation to a later version, see "Upgrading Data Replication" on page 20.

Note: To install Data Replication on Windows, you must log in to the system as the Administrator user or log in with a user ID that belongs to the Administrator group.

1. Transfer the platform-specific installation file that you received from Informatica to a directory on the system.

   The installation file is a .zip file on Windows or a tar.gz file on Linux or UNIX.

   Tip: Informatica recommends that you retain this file in case you need to reinstall the product.
2. Extract or decompress the installation file into a directory that is owned by the operating system user:
   - On Windows, extract the installation .zip file.
   - On Linux or UNIX, decompress the tar.gz file or .tar file.
3. Verify that the expected Data Replication files and directories are present.

   For a list of installed subdirectories, programs, and files, see "Appendix B: Data Replication Files and Subdirectories" in the *Informatica Data Replication User Guide*.
4. Define the DBSYNC_HOME environment variable to identify the Data Replication installation root directory. For more information, see "Defining the DBSYNC_HOME Environment Variable" on page 13.
5. To quickly verify that Data Replication was installed, perform one of the following actions:
   - On Windows, double-click idr_console.cmd in the root installation directory to start the Data Replication Console. You can view the Data Replication Console version and build number on the splash screen and by clicking **Help** > **About**.

     Note: You can run the Data Replication Console only on 64-bit Windows systems.
   - On Linux or UNIX, run server_manager.sh to start the Server Manager. The Data Replication Server Manager version and build number is displayed in the command window.

Post-Installation Tasks

After you complete installation and before you begin creating replication configurations, perform the following tasks:

- Verify that a supported JRE version is installed. See "Installing the Java Runtime Environment" on page 12.
- Define the DBSYNC_HOME environment variable. See "Defining the DBSYNC_HOME Environment Variable" on page 13.
- On a Linux or UNIX system, add the DBSYNC_HOME libraries to the library path environment variable. See "Adding DBSYNC_HOME Libraries to the Library Path Environment Variable on Linux and UNIX" on page 13.
- Verify the environment variables. See "Verifying the Environment Variables on Linux and UNIX" on page 13.
- Set Limits for System Resources on Linux and UNIX. See "Setting Limits for System Resources on Linux and UNIX" on page 14.
- Enter the license key. See "Entering the License Key" on page 14.
- On an AIX system, set the AIXTHREAD_STK environment variable to 8000000. InitialSync requires this setting for multithreaded processing.
- Verify that the JDBC drivers that are required by the Data Replication Console and Replication Configuration CLI are installed in the *DataReplication_installation*/lib subdirectory. See "JDBC Drivers" on page 14.
- Verify that you have the correct ODBC drivers to connect to the source or target database. See "ODBC Drivers" on page 16.
- On a Linux or UNIX system, configuring DataDirect ODBC drivers. See "Configuring the DataDirect ODBC Drivers and Driver Manager" on page 18.
- Prepare the source and target systems. See the *Informatica Data Replication User Guide*.

Installing the Java Runtime Environment

The Data Replication Console, the Replication Configuration Command Line Interface (CLI), and the Applier for Apache Kafka, Cloudera and Hortonworks targets require the Java Runtime Environment (JRE).

1. Verify that the Java Runtime Environment (JRE) 1.7, 1.8, 9.0.1, or 10 is installed. Use the 64-bit JRE version.

   The Java Virtual Machine (JVM) allocates contiguous memory for the Java heap. On Windows, the idr_console.cmd script runs the JVM with the -Xmx1024m option to specify the maximum Java heap size of 1024 MB for the Data Replication Console. Ensure that the system where you plan to run the Data Replication Console has 1024 MB of contiguous memory space.
2. If you plan to encrypt the database passwords that you specify for source and target database connections, install Java Cryptography Extension (JCE). Perform the following substeps:
   a. Download JCE Unlimited Strength Jurisdiction Policy Files 7 from the Oracle website:

      http://www.oracle.com/technetwork/java/javase/downloads/jce-7-download-432124.html
   b. Decompress the .zip file into the *Java Runtime Environment*/jre/lib/security subdirectory.
3. Define the JRE_HOME environment variable to point to the JRE base directory.
4. Add the %JRE_HOME%\bin directory to the beginning of the PATH environment variable.

To verify that the correct JRE version is installed, issue the following command from the command line:

```
java -version
```

Defining the DBSYNC_HOME Environment Variable

On each system with a Data Replication installation, define the DBSYNC_HOME environment variable in the user profile to identify the Data Replication installation root directory.

Use the following syntax to define a DBSYNC_HOME system environment variable that points to the Data Replication installation directory:

```
DBSYNC_HOME=DataReplication_install_directory
```

This environment variable pertains to all of the operating system types that Data Replication supports.

Adding DBSYNC_HOME Libraries to the Library Path Environment Variable on Linux and UNIX

On Linux and UNIX, add the following DBSYNC_HOME libraries in the library path environment variable to define search paths that the dynamic linker uses to find required shared libraries:

- DBSYNC_HOME
- DBSYNC_HOME/support
- DBSYNC_HOME/dd/lib

Use the library path environment variable for your operating system:

- LD_LIBRARY_PATH on Linux
- LD_LIBRARY_PATH_64 on Solaris
- LIBPATH on AIX
- SHLIB_PATH on HP-UX

For example:

```
LIBPATH=$DBSYNC_HOME:$DBSYNC_HOME/support:$DBSYNC_HOME/dd/lib:$ORACLE_HOME/lib:$LIBPATH
```

Verifying the Environment Variables on Linux and UNIX

To verify that you correctly defined the library path, installation directory, and ODBC system information environment variables on Linux and UNIX, use the ldd utility to print the shared libraries required for the InitialSync, Extractor, and Applier executables.

For example, use the following commands to run the ldd utility for each executable that works with Teradata sources or targets:

```
ldd dbsync_extract
ldd dbsync_initial_td
ldd dbsync_apply_td
```

The ldd utility prints the shared library paths for each executable. If any shared libraries are listed as not found, an environment variable is either not defined or not defined incorrectly. In this situation, verify that all of the environment variables are defined correctly, and then run the ldd utility again.

For details about the executable names to use with the ldd utility, see GUID-268E375B-97A0-45BD-9EFA-229D9B8DF9A1 and "Configuring the DataDirect ODBC Drivers and Driver Manager" on page 18.

Setting Limits for System Resources on Linux and UNIX

On Linux and UNIX, you must configure the system resources that are available to Data Replication processes, including InitialSync, Extractor, Applier, and Server Manager processes. Adequate system resources are required to replicate changes accurately and provide debug information in case of failure.

Use the ulimit command to set maximum limits for the system resource settings.

Tip: To get a list of limits for all of the system resource settings and corresponding options for the ulimit command, run the following command:

```
ulimit -a
```

1. Set the maximum size for core files. Use the following command:

   ```
   ulimit -c unlimited
   ```

2. Set the maximum size for a process data segment. Use the following command:

   ```
   ulimit -d unlimited
   ```

3. Set the maximum size for the files that a process can create. Use the following command:

   ```
   ulimit -f unlimited
   ```

4. Set the maximum stack size. Use the following command:

   ```
   ulimit -s unlimited
   ```

5. Set the maximum amount of CPU time. Use the following command:

   ```
   ulimit -t unlimited
   ```

Entering the License Key

Verify that you have a valid license key for Data Replication and then enter the key value in the key file.

If you download the installation files from the Informatica Electronic Software Download (ESD) site, the license key is delivered in an email message from Informatica.

1. To enter the license key in the DataReplication.key file, use one of the following methods:
   - Enter the license key value directly in the DataReplication.key file, which is located in the top-level Data Replication installation directory. Replace the entire contents of the file with the key value.
   - Rename the .lic file that contains the license key to DataReplication.key and then copy the file to the top-level Data Replication installation directory.
2. If you run Data Replication on multiple systems, copy the DataReplication.key file to all of the Data Replication instances.

If you have questions regarding the license key, contact Informatica Global Customer Support.

JDBC Drivers

The Data Replication Console and Replication Configuration Command Line Interface use JDBC drivers to connect to databases. Data Replication does not use JDBC drivers for Apache Kafka, Flat File, Cloudera, and Hortonworks targets.

Data Replication provides some JDBC drivers in the *DataReplication_installation*/lib subdirectory. For Netezza, Teradata, Vertica, and earlier versions of MySQL,you must download the appropriate JDBC drivers.

The following table describes the JDBC drivers and environment variables that the Data Replication Console requires:

Database	Driver	Environment Variable	Driver Provided by Data Replication?
Amazon Redshift	PostgreSQL JDBC driver	-	Yes, in the lib subdirectory
DB2 for Linux, UNIX, and Windows	DataDirect DB2 JDBC driver	-	Yes, in the lib subdirectory
Greenplum	PostgreSQL JDBC driver	-	Yes, in the lib subdirectory
Microsoft SQL Server	DataDirect SQL Server JDBC driver	-	Yes, in the lib subdirectory
MySQL	DataDirect MySQL JDBC driver	-	Yes, in the lib subdirectory
Netezza	Netezza JDBC driver, nzjdbc.jar	NETEZZA_JDBC_DRIVER_LIB	No. Download a Netezza JDBC driver version that is compatible with your database version from the Netezza website. Copy the files to the *DataReplication_installation*/lib subdirectory.
Oracle	Oracle JDBC driver	-	Yes, in the lib subdirectory
PostgreSQL	PostgreSQL JDBC driver	-	Yes, in the lib subdirectory
Teradata	For Teradata 13.10, the JDBC driver files: - terajdbc4.jar - tdgssconfig.jar - tdgssjava.jar For Teradata 14.0 and 14.10, the JDBC driver files: - terajdbc4.jar - tdgssconfig.jar	TERADATA_JDBC_DRIVER_LIB	No. Download a Teradata JDBC driver version that is compatible with your database version from the Teradata website. Copy the files to the *DataReplication_installation*/lib subdirectory. **Important:** If you use Teradata version 14.10, download version 14.10.00.44 of the JDBC driver files or a later version of the files.
Vertica	Vertica JDBC driver, vertica-jdk5-6.1.2-0.jar	VERTICA_JDBC_DRIVER_LIB	No. Download the Vertica JDBC driver version that is compatible with your database version from the Vertica website. Copy the files to the *DataReplication_installation*/lib subdirectory.

For Netezza, Teradata, and Vertica targets, verify that the JDBC drivers you use have the .jar file names that are specified in the table. If not, open the following files in a text editor and edit the environment variable values to specify the JDBC drivers that are in use:

- idr_console.cmd
- idr_cmd_line.sh or idr_cmd_line.cmd

- idrcmd.sh or idrcmd.bat

ODBC Drivers

Data Replication uses ODBC drivers to connect to most targets to apply or synchronize change data. Data Replication also uses ODBC drivers to connect to a few types of sources to read data for extraction processing.

On Windows, download the appropriate ODBC driver from the database vendor if you do not already have it, or ensure that the required database client is installed. On Linux or UNIX, use one of the DataDirect ODBC drivers that Data Replication provides.

Data Replication provides DataDirect ODBC drivers in the *DataReplication_installation*/dd/lib subdirectory for the following databases:

- DB2 for Linux, UNIX, and Windows
- Greenplum
- Microsoft SQL Server
- MySQL
- PostgreSQL

For Netezza and Vertica on Linux or UNIX, download the DataDirect ODBC driver from the database vendor. Put the DataDirect ODBC driver files in the *DataReplication_installation*/dd/lib subdirectory.

Data Replication does not use ODBC drivers for Apache Kafka, Cloudera, Flat File, and Hortonworks targets.

The following table describes the Data Replication requirements for ODBC drivers, including whether a driver is required on the source, target, or both:

Database	Source	Target	Comment
Amazon Redshift	-	Yes	On Windows, install the PostgreSQL ODBC driver. On Linux, install the Amazon Redshift ODBC driver. For more information, see the "Preparing Amazon Redshift Target Systems" topic in the *Informatica Data Replication User Guide*.
DB2 for Linux, UNIX, and Windows	Yes, for InitialSync	Yes, for InitialSync	For DB2 sources and targets, ensure that the DB2 Client, DB2 Runtime Client, or DB2 Connect is installed. The Extractor uses the DB2 Call Level Interface (CLI), which is provided with the clients, to access the database for transactional replication. To run an InitialSync task on Windows, use the IBM DB2 ODBC driver. This driver is usually delivered with the DB2 Client. If necessary, you can manually download it. To run InitialSync on Linux and UNIX, use the DataDirect ODBC driver for DB2 for Linux, UNIX, and Windows that Data Replication provides.
Greenplum	-	Yes	On Windows, install the Greenplum ODBC driver (preferred) or the PostgreSQL ODBC driver for Greenplum targets. On Linux and UNIX, use the DataDirect ODBC driver for Greenplum that Data Replication provides.

Database	Source	Target	Comment
MemSQL	-	Yes	Install the MySQL ANSI ODBC driver 5.3.11.
Microsoft SQL Server	Yes, on Linux and UNIX	Yes, if the Data Replication apply process that accesses the SQL Server target runs on Linux and UNIX	On Windows, ensure that the SQL Server Native Client is installed. On Linux and UNIX, use the DataDirect ODBC driver for Microsoft SQL Server that Data Replication provides.
MySQL	Yes	Yes	On Windows, InitialSync uses the MySQL ODBC driver to load data. On Linux, and on UNIX (targets only), InitialSync uses the DataDirect ODBC driver for MySQL that Data Replication provides.
Netezza	-	Yes	On Windows, install the Netezza ODBC driver for Netezza targets. On Linux and UNIX, install the DataDirect ODBC driver for Netezza targets.
Oracle	No	No	For Oracle sources and targets, ensure that the Oracle Client is installed. InitialSync uses the OCI library. For replication from Oracle sources to Oracle targets, InitialSync can alternatively use DBLinks to unload data.
PostgreSQL	-	Yes	On Windows, install the PostgreSQL ODBC driver for PostgreSQL targets. On Linux and UNIX, use the DataDirect ODBC driver for PostgreSQL that Data Replication provides.
Teradata	-	Yes	For Teradata targets, install the TPT libraries that include the Teradata ODBC driver. For more information, see the "Preparing Teradata Target Systems" topic in the *Informatica Data Replication User Guide*.
Vertica	-	Yes	On Windows, install the Vertica ODBC driver for Vertica targets. On Linux and UNIX, install the DataDirect ODBC driver for Vertica targets. On HP-UX, install version 6.1.2 or later of the DataDirect ODBC driver for Vertica.

Important: Use the driver or client version that matches your database version.

On Windows, if you installed multiple ODBC driver versions on one system, you must specify the name of the ODBC driver that you want to use. Specify this name in the **Custom connection string** field for the source or target database in the Data Replication Console.

On Linux and UNIX, verify the settings for the DataDirect ODBC driver in the *DataReplication_installation*/dd/odbcinst.ini file.

Configuring the DataDirect ODBC Drivers and Driver Manager

On Linux and UNIX, you must configure the DataDirect ODBC drivers and driver manager that Data Replication requires to connect to sources and targets.

The DataDirect ODBC driver manager provides an ODBC API that Data Replication uses to communicate with the ODBC drivers.

1. You will need to open the Informatica Data Replication Console at a later time and set the global.use_data_direct parameter to 1 to indicate that Data Replication will use the DataDirect ODBC driver manager. On the **Runtime Settings** tab > **Advanced Settings** view, the global.use_data_direct parameter is not visible. To ensure that this parameter is set to 1:
   a. Click **New**.
   b. Enter the global.use_data_direct parameter name in the **Input** dialog box. Click **OK**.
   c. Enter the parameter value in the **Input** dialog box. Click **OK**.

   The global.use_data_direct parameter appears in the User Defined list on the **Advanced Settings** view.
2. Verify the driver settings for your sources and targets in the *DataReplication_installation*/dd/odbcinst.ini file, including the path to the DataDirect ODBC driver. This file contains driver settings for all of the supported source and target database types.
3. Optionally, set environment variables for the DataDirect ODBC driver and driver manager prior to starting the Server Manager. The Server Manager script does not set the DataDirect ODBC driver environment variables if you have already set them manually.

   Alternatively, wait until you start the Server Manager. The server_manager.sh script sets all of the required environment variables for the DataDirect ODBC driver.
   a. Define the DD_INSTALLDIR environment variable. This variable points to the *DataReplication_installation*/dd directory.
   b. Add DataDirect driver manager libraries to the system path.

      Data Replication provides the DataDirect driver manager libraries in the *DataReplication_installation*/dd/lib directory. Use a system path environment variable to point to the directory that contains the DataDirect driver libraries.
      The following table describes system path environment variables for the supported Linux and UNIX operating systems:

Operating System	Environment Variable
AIX	LIBPATH
HP-UX	SHLIB_PATH
Linux	LD_LIBRARY_PATH
Solaris	LD_LIBRARY_PATH_64

   c. Set ODBCINST to point to the *DataReplication_installation*/dd/odbcinst.ini file. For example:

```
export ODBCINST=$DD_INSTALLDIR/odbcinst.ini
```

License Expiration

A month prior to the license expiration date, the Server Manager starts sending warning messages, which you can see in the Server Manager log and in the Event Viewer in the Data Replication Console.

When the license expires, the Server Manager stops all of the running replication tasks and ends.

To extend the license, contact Informatica Global Customer Support.

After you extend the license and copy the new DataReplication.key file to all of the Data Replication instances, you can start the Server Manager and then start your replication tasks and schedules in the Data Replication Console.

CHAPTER 2

Upgrading Data Replication

This chapter includes the following topics:

- Overview, 20
- Upgrading Data Replication, 20

Overview

If you are upgrading from Data Replication 9.6.2 or a later 9.6.*x* release to the current release, extract or decompress the Data Replication installation file for the current release into the existing Data Replication installation directory.

If you run Data Replication on multiple systems, upgrade Data Replication and all Data Replication Console installations on all of the systems. Do not run the current release on one system and run an earlier Data Replication release on another system in the same topology. For example, if you run the current release of a Server Manager with an earlier release of a Data Replication Console, the Console will not be able to connect to the Server Manager.

If you are upgrading from 9.6.1 or earlier, contact Informatica Global Customer Support for assistance.

Upgrading Data Replication

Upgrade Data Replication on each system where you run a Data Replication component such as the Extractor, Applier, Server Manager, InitialSync, and Data Replication Console. Use this procedure for an existing installation.

1. Download the platform-specific Data Replication installation file for the current release to a directory on the local system.

   Tip: Retain the installation file so that you can use it to upgrade other Data Replication instances or redo an upgrade if necessary.
2. If you use DB2 for Linux, UNIX, and Windows or Microsoft SQL Server sources, perform the following substeps for each configuration with a DB2 or SQL Server source:
   a. Ensure that the Extractor completed processing the transaction logs.
   b. Ensure that the source database does not have open transactions.

Tip: To view the list of transactions that were open in the source database at the time of the last Extractor checkpoint, on the **Server Manager** tab > **Configs** view, select a configuration and then click the **Show Transactions** icon button on the Replication Configurations toolbar.

c. Ensure that the Applier applied change data from all of the intermediate files.

Tip: To view the list of intermediate files and their processing statuses, on the **Server Manager** tab > **Configs** view, select a configuration for which to view intermediate files. Then click the **Intermediate Files** icon button on the Replication Configurations toolbar.

3. Stop all replication tasks and schedules.
4. Stop all Server Manager servers that are running under the earlier Data Replication version.

   Important: Stop Server Manager Main servers first. Then stop Server Manager subservers.

   If you are running the Server Manager as a service or a daemon, use the following commands to stop and remove the service or daemon:

   - On Windows:

     ```
     cd %DBSYNC_HOME%
     server_manager.exe RUN_AS_SERVICE -k
     server_manager.exe RUN_AS_SERVICE -r
     ```

   - On Linux:

     ```
     cd $DBSYNC_HOME
     ./server_manager.sh stop
     ```

     Verify that the Server Manager daemon is not running. For example, use the following command:

     ```
     ps -eaf | grep server_manager
     ```

5. If any existing configurations use source or target connections that are not saved in the **Server Manager** > **Connections** view of the Data Replication Console, Informatica recommends that the idradmin user save these connections with unique names before upgrading the Data Replication installation. Otherwise, Data Replication 9.7.0 will automatically create separate named connections for each configuration, even if multiple configurations use the same connection details. The names of these connections will have the format <*configuration_name*>_source_connection or <*configuration_name*>_target_connection.

   Note: Connection names can contain the digits 0-9, Latin letters A-Z and a-z, and the underscore (_) character. The Data Replication Console truncates connection names that are longer than 120 characters.

6. Back up the existing Data Replication installation, including the intermediate files and replication configurations.
7. Delete the support and lib subdirectories from the existing Data Replication installation.
8. Delete the log4j.xml file from the uiconf subdirectory in the existing Data Replication installation.
9. Delete the state.dat file from the existing Data Replication installation.
10. Extract or decompress the installation file for the current release into a temporary directory:
    - On Linux, decompress the DataReplication_*v_r_m*.linux.x86_64.tar.gz file.
    - On Windows, extract the DataReplication_*v_r_m*.windows.64bit.zip file.
11. Copy the extracted files from the temporary directory into the existing Data Replication installation directory.

    Confirm that the files for the previous version in the Data Replication installation directory were overwritten.

12. If you have a new license key file, copy the new DataReplication.key file to the Data Replication installation directory. Confirm that the old license key file was overwritten.

13. If you previously installed JDBC drivers in the lib subdirectory, copy these files from the backup copy of the Data Replication installation to the lib subdirectory in the current Data Replication installation directory.
14. If you previously replicated change data to a Hadoop Distributed File System (HDFS), perform the following substeps:
    a. Download the hadoop_libs.zip file that Data Replication provides. This file includes the .jar files that Data Replication requires for Cloudera and Hortonworks targets.
    b. Extract the hadoop_libs.zip file into the *DataReplication_installation* directory.
    c. If the lib/hadoop subdirectory in the backup copy of the Data Replication installation includes .xml files, copy these files to the lib/hadoop subdirectory in the current Data Replication installation.
15. If you previously changed or added parameters in the *DataReplication_installation*/uiconf/default.cfg file, retain the existing default.cfg file in the current Data Replication installation. If you did not previously change the default.cfg file, you can delete this file from the current uiconf subdirectory.

    Note: Beginning with version 9.7.0, Data Replication no longer installs this file. If Informatica Global Customer Support directs you to create a default.cfg file, retain the file each time you upgrade.
16. If you previously defined post-apply SQL statements or stored procedure calls in the *DataReplication_installation*/uiconf/afterapply.xsl file, copy the changes from the backup copy of the file to the afterapply.xsl file in the current Data Replication installation.
17. If you previously customized the DataTypes.xml or transform.xsl file in the *DataReplication_installation*/uiconf directory at the direction of Informatica Global Customer Support or product specialists, copy the changes from the backup copy of the file to the appropriate sections of the new file in the uiconf directory.
18. Verify that the DBSYNC_HOME environment variable still points to the Data Replication installation.
19. For the systems where the Data Replication Console and the Applier for Cloudera and Hortonworks targets run, verify that a supported version of JRE is installed. For more information about the supported JRE versions, see "Installing the Java Runtime Environment" on page 12. To determine the installed version, issue the following command from the command line:

    ```
    java -version
    ```
20. On Windows, verify that the Microsoft Visual C++ 2008 Redistributable Package (x64) is installed.
21. If you plan to encrypt the database passwords that you specify for source and target database connections, install the Java Cryptography Extension (JCE). Perform the following substeps:
    a. Download JCE Unlimited Strength Jurisdiction Policy Files 7 from the Oracle web site:

       http://www.oracle.com/technetwork/java/javase/downloads/jce-7-download-432124.html
    b. Decompress the .zip file into the *%JRE_HOME%*\lib\security subdirectory.
22. Restart all of the Server Manager servers.

    Important: Start Server Manager subservers first. Then start Server Manager Main servers.

    If you are running a Server Manager instance as a service or a daemon, use the following commands to restart the service or daemon:

    - On Windows:

      ```
      cd %DBSYNC_HOME%
      server_manager.exe RUN_AS_SERVICE -i
      server_manager.exe RUN_AS_SERVICE -s
      ```
    - On Linux:

      ```
      cd $DBSYNC_HOME
      ./server_manager.sh
      ```
23. From the Data Replication Console, connect to the Server Manager Main server.

If the Server Manager Main server uses HTTPS connections, the host name or IP address in the cert.pem file must match the host name or IP address entered in the Console to connect to the Server Manager Main server. If the host names or IP addresses do not match, regenerate the cert.pem file or edit the information in the **Host** field on the **Server Manager** tab > **Configs** view before connecting to the Server Manager Main server.

24. Restart all of the replication tasks and schedules in the current Data Replication installation.

INDEX

D

DataDirect ODBC drivers
  configuring DataDirect ODBC drivers and driver manager 18
DBSYNC_HOME environment variable 13

E

environment variables
  verifying on Linux and UNIX 13

I

installation 11

J

Java Runtime Environment (JRE) 12
JDBC drivers 14

L

license key
  entering the license key 14
  expiration 19

O

ODBC drivers
  configuring DataDirect ODBC drivers and driver manager 18
  driver requirements by database type 16

S

source databases
  supported database versions 9
system requirements
  AIX 7
  HP-UX 7
  Linux 8
  Windows 9
system resources
  setting limits for replication processes 14

T

target databases
  supported database versions 10

U

upgrade 20